

Arshad Murtaza

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PROFESSIONAL SUMMARY

Aspiring AI Engineer with hands-on experience building and deploying production-grade AI systems, including RAG pipelines, LLM-powered applications, and end-to-end MLOps workflows. Proficient in Python, LangChain, FastAPI, and AWS, with a strong foundation in Deep Learning, NLP, and Generative AI. Demonstrated ability to take projects from model development to cloud deployment using Docker, GitHub Actions CI/CD, and scalable AWS infrastructure (ECS Fargate, ECR). Seeking to contribute to innovative AI engineering teams building real-world intelligent systems.

SKILLS

Languages & Tools: Python, SQL, Excel, Git, Docker, AWS (ECS, ECR), GitHub Actions (CI/CD), Power BI

AI & Machine Learning: Machine Learning, Deep Learning, NLP, Transformers, Generative AI, Time-Series Analysis

Frameworks & Libraries: LangChain, FastAPI, TensorFlow/Keras, MLOps, LLMops, EDA

PROJECTS

Document Portal 🔗

Mar 2026 - Mar 2026

- Built a production-grade AI Document Portal using FastAPI, LangChain, and FAISS — integrating Groq/Gemini LLMs to deliver automated PDF summarization, document comparison, and a conversational RAG-powered Q&A interface for natural-language document querying.
- Architected an end-to-end MLOps pipeline by containerizing with Docker, provisioning AWS infrastructure (ECS Fargate + ECR) via CloudFormation, and automating zero-downtime CI/CD through GitHub Actions with secrets secured via AWS Secrets Manager.

Plant Disease Classification using CNN 🔗

Nov 2025 - Nov 2025

- Developed CNN architecture from scratch for automated plant disease detection classifying 38 disease categories from leaf images, achieving 88% accuracy using TensorFlow/Keras with training on Google Colab (T4 GPU).
- Deployed interactive web application using Streamlit for real-time disease prediction and inference, enabling end-to-end machine learning workflow from model development to production deployment.

Customer Segmentation using K-Means Clustering 🔗

Oct 2025 - Oct 2025

- Performed end-to-end customer segmentation on UCI Online Retail dataset using K-Means clustering with RFM (Recency, Frequency, Monetary) feature engineering, identifying 4 distinct behavioral segments to enable targeted marketing strategies.
- Deployed end-to-end ML pipeline with optimal cluster selection using Elbow Method and Silhouette Score, PCA dimensionality reduction achieving 85.8% variance retention, and interactive Streamlit dashboard for real-time customer segment visualization and business insights.

Credit Card Fraud Detection using Machine Learning 🔗

Oct 2025 - Oct 2025

- Developed fraud detection system analyzing credit card transactions using Logistic Regression, Random Forest, and XGBoost classifiers, achieving 87.1% precision and 82.7% recall with Random Forest on highly imbalanced Kaggle dataset.
- Implemented SMOTE oversampling technique to handle extreme class imbalance and conducted comprehensive model evaluation using precision-recall metrics, demonstrating strong fraud detection capability with minimal false positives for real-world deployment.

EDUCATION

IIT Delhi

B.Tech in Civil Engineering

Aug 2019 - Jun 2023

ACHIEVEMENTS & CERTIFICATIONS

- Data Science and Machine Learning with GenAI 🔗